

BIRA-MPPI: Belief-Informed Responsiveness-Aware Planning for Robot Navigation in Dynamic Crowds

Abstract—Safe crowd navigation requires anticipating pedestrian reactions to robot motion under uncertain individual responsiveness. We propose a belief-informed responsiveness-aware framework based on Model Predictive Path Integral (MPPI) control (BIRA-MPPI). A Virtual Risk Corridor (VRC) models the robot’s trajectory-dependent influence, scaled by pedestrian-specific latent responsiveness gains. Beliefs over these gains are updated online from observed interactions under a variational free-energy formulation. Propagating sampled gains through VRC dynamics yields pedestrian trajectory distributions conditioned on each robot candidate. The planner combines expected cost with conditional value-at-risk and independently refines five motion modes to preserve multimodality, recomputing predictions and risks before final selection. Closed-loop SocNavGym experiments with ten Optimal Reciprocal Collision Avoidance (ORCA) pedestrians cover uncooperative, probabilistic, and cooperative response regimes, using 50 matched scenarios per regime. BIRA-MPPI achieves higher success rates and lower collision rates than constant-velocity and fixed-responsiveness VRC baselines across all regimes. The source code is available at <https://bira-mpai.github.io>.

I. INTRODUCTION

Mobile robots navigating among pedestrians must anticipate human motion while making safe progress toward their goals [1], [2]. Robot motion can influence pedestrian behavior, but individual reactions vary: real-world interaction data include both avoidance and neutral responses [3]. A crossing maneuver may be safe if a pedestrian yields yet unsafe if they continue without reacting. Conversely, ignoring possible cooperation can exclude feasible passages and contribute to the freezing robot problem [4]. Navigation therefore requires accounting for both the robot’s influence and uncertainty in pedestrian responsiveness.

Coupled prediction and planning evaluate surrounding-agent motion conditioned on candidate robot actions [5], [6], [7], but this conditioning alone does not quantify uncertainty in how strongly a pedestrian will respond. Online behavioral inference can adapt response estimates from observations [8], [9], yet a point estimate may conceal weaker responses that make a plan unsafe. Propagating this uncertainty into risk-sensitive planning allows expected performance to be balanced against adverse outcomes [10], [11]. We therefore ask: *how can online beliefs about pedestrian-specific responsiveness inform the evaluation and refinement of alternative robot trajectories?*

We propose a belief-informed responsiveness-aware framework based on MPPI [12], as shown in Fig. 1. A VRC represents the robot’s trajectory-dependent influence through motion-adaptive ellipses and a finite-preview interaction force. Pedestrian-specific effective gains scale this force,

with beliefs updated online through approximate Bayesian inference under a variational free-energy formulation [13]. Sampling these gains yields pedestrian trajectory distributions conditioned on each robot candidate, without requiring a pretrained predictor. The uncertainty modeled here is limited to responsiveness relative to the VRC.

The main contributions are: (i) **Online responsiveness inference**. We maintain beliefs over pedestrian-specific response gains and update them recursively from observed robot–pedestrian interactions. (ii) **Action-conditioned distributional prediction**. We propagate these beliefs through VRC dynamics to generate a distribution of pedestrian responses for each robot candidate. (iii) **Belief-aware risk-sensitive planning**. We combine expected cost and conditional value-at-risk in MPPI, independently refining motion modes and re-evaluating their interaction outcomes before selection.

II. RELATED WORK

Interaction-Aware Prediction and Planning for Crowd Navigation. Interaction-aware navigation accounts for the dependence of pedestrian motion on robot actions. Classical approaches describe local avoidance through social forces [14], [15] or reciprocal velocity constraints [16]. Moving beyond local avoidance, Trautman and Krause [4] introduced interacting Gaussian processes to jointly model robot and pedestrian trajectories, addressing the freezing robot problem caused by neglecting cooperative interactions.

Recent approaches incorporate these dependencies into predictive planning through explicit interaction models or learned predictors. SICNav [17] embeds ORCA-based pedestrian responses within bilevel Model Predictive Control (MPC), while SFM-NMPC [18] incorporates social-force dynamics directly into the optimization loop. Learning-based formulations include multimodal human responses conditioned on candidate robot actions [19] and MATS [20], which represents interacting agent dynamics as mixtures of affine time-varying systems. For crowd navigation, Le et al. [5] couple Social-LSTM prediction with MPC through iterative best-response updates. SICNav-Diffusion [6] combines joint diffusion predictions with ORCA-based safety refinement, coupling the refined human trajectories with the robot plan. In the related autonomous-driving setting, IANN-MPPI [7] generates vehicle predictions conditioned on each sampled ego-control sequence, enabling candidate-specific interaction evaluation.

Pedestrian Response Uncertainty and Online Inference. Probabilistic forecasters such as Trajectron++ [21] represent

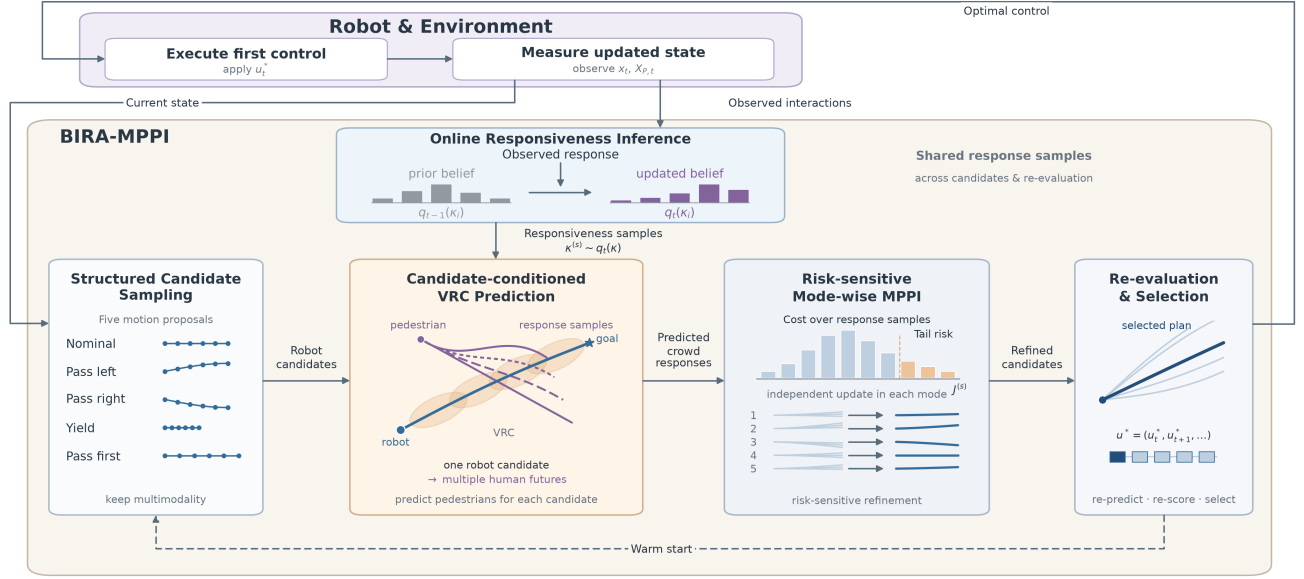


Fig. 1. BIRA-MPPI framework. Online responsiveness beliefs generate VRC-conditioned pedestrian predictions for each robot candidate. Risk-sensitive MPPI refines motion modes independently, then recomputes pedestrian predictions and risks for the refined trajectories before selecting the final plan.

multiple plausible futures, while DD-MDN [22] additionally addresses predictive uncertainty calibration. Complementary approaches adapt behavioral models from online observations. SocioSense [23] combines Bayesian learning with personality-based modeling to estimate time-varying pedestrian behaviors and adapt prediction and interpersonal spacing. Fisac et al. [8] infer model confidence online, broadening human-motion predictions when observations deviate from the assumed model.

Beyond passive observation, Sadigh et al. [24] select robot actions that reveal human internal states. Nam and Kwon [9] jointly infer human characteristics and a continuous rationality parameter, incorporating future belief updates into sampling-based stochastic MPC. These approaches demonstrate that behavioral inference can be integrated with planning rather than performed solely as an independent prediction stage.

Within a free-energy-inspired framework, Dang et al. [25] recursively update Gaussian beliefs over velocity. They use the Kullback-Leibler (KL) divergence between predictions and observations as a proxy for cognitive uncertainty to modulate social forces and graph-based vehicle decision-making. Our approach instead targets uncertainty in each pedestrian’s effective response strength relative to the VRC model. A variational Bayesian formulation supports online gain estimation from observed interactions, and the resulting beliefs are propagated through robot-conditioned prediction into risk-sensitive MPPI.

III. PRELIMINARIES: VARIATIONAL BAYESIAN INFERENCE

Let z denote a latent variable, $\mathbf{y}$ an observation, and $p^-(z)$ the prior belief before incorporating $\mathbf{y}$. Variational Bayesian inference approximates the posterior $p(z | \mathbf{y})$ with a tractable

belief $q(z)$ by minimizing the variational free energy [13]

$$\begin{aligned} \mathcal{F}[q] &= -\mathbb{E}_q[\log p(\mathbf{y} | z)] + D_{\text{KL}}(q(z) \| p^-(z)) \\ &= D_{\text{KL}}(q(z) \| p(z | \mathbf{y})) - \log p(\mathbf{y}). \end{aligned} \quad (1)$$

Since the log evidence $\log p(\mathbf{y})$ is independent of q , minimizing $\mathcal{F}[q]$ drives $q(z)$ toward the Bayesian posterior. We later apply this principle to recursively infer pedestrian responsiveness from observed robot–pedestrian interactions.

IV. PROBLEM FORMULATION

We consider a robot navigating toward a goal $\mathbf{g}$ among N pedestrians whose responses to its motion are heterogeneous and uncertain. The objective is to achieve safe and efficient navigation by accounting for both trajectory-dependent pedestrian reactions and uncertainty in their response strength.

At each replanning cycle t , the robot plans over T prediction steps. Let $\mathbf{x}_k = (x_k, y_k, \theta_k)^\top$ and $\mathbf{u}_k = (v_k, \omega_k)^\top$ denote its state and control input at prediction step k , with the conditioning on t omitted for compactness. The discrete unicycle dynamics are

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \Delta t \begin{bmatrix} v_k \cos \theta_k \\ v_k \sin \theta_k \\ \omega_k \end{bmatrix}, \quad (2)$$

where Δt is the time interval. Starting from the observed robot state $\mathbf{x}_0$, a control sequence $\mathbf{u} = [\mathbf{u}_0, \dots, \mathbf{u}_{T-1}]$ generates the trajectory $\mathbf{x} = [\mathbf{x}_0, \dots, \mathbf{x}_T]$.

The robot observes the current pedestrian positions and velocities, collected in $\mathbf{X}_{P,0}$. Each pedestrian i is associated with a latent responsiveness gain $\kappa_i \in [0, \kappa_{\max}]$, which scales its response to the VRC-induced interaction force. The robot maintains a belief $q_t(\kappa)$ over $\kappa = (\kappa_1, \dots, \kappa_N)$ and updates it online from observed pedestrian responses.

For a sampled $\kappa \sim q_t$, the corresponding crowd prediction is

$$\hat{\mathbf{X}}_P = \mathcal{G}(\mathbf{X}_{P,0}, \mathbf{x}, \mathbf{u}, \kappa), \quad (3)$$

where $\mathcal{G}$ denotes the VRC-based pedestrian response model. The sampled gain is held fixed within each rollout. Consequently, each candidate robot trajectory induces a distribution of possible pedestrian responses, $p(\hat{\mathbf{X}}_P | \mathbf{x}, \mathbf{u}, q_t)$.

The finite-horizon planning problem is formulated as

$$\mathbf{u}_t^* = \arg \min_{\mathbf{u} \in \mathcal{U}^T} \mathcal{J}(\mathbf{u}; q_t), \quad (4)$$

subject to (2), where $\mathcal{U}$ denotes the admissible control set. The objective evaluates each candidate under the belief-induced pedestrian trajectory distribution, accounting for both expected navigation performance and interaction risk.

The main challenges are to infer pedestrian-specific responsiveness from ongoing interactions and to account for the resulting action-conditioned predictive uncertainty in planning. We address these challenges with BIRA-MPPI, which integrates online Bayesian belief updates, VRC-based distributional prediction, and risk-sensitive trajectory optimization within a receding-horizon framework.

V. PEDESTRIAN PREDICTION VIA VIRTUAL RISK CORRIDORS

A. Virtual Risk Corridor Construction

For each candidate robot trajectory, the VRC represents its motion-dependent influence through a time-indexed sequence of ellipses. Let $r_0 = r_R + r_P + d_{\text{personal}}$, where r_R and r_P are the robot and pedestrian radii, and d_{personal} is the additional social clearance. The longitudinal and lateral semi-axes at step k are

$$\begin{aligned} a_k &= r_0 + \tau_f |v_k| + \frac{v_k^2}{2a_{\text{brake}}}, \\ b_k &= r_0 + \tau_l |v_k| + c_\omega |\omega_k|. \end{aligned} \quad (5)$$

Here, $\tau_f, \tau_l > 0$ scale the speed-dependent margins, $a_{\text{brake}} > 0$ is the assumed braking deceleration, and $c_\omega > 0$ controls turning-induced lateral expansion. The longitudinal extent incorporates speed-dependent spacing [26], [27] and braking distance.

To represent directional asymmetry [28], the ellipse center is shifted along the robot's direction of motion:

$$\mathbf{c}_k = \mathbf{p}_{R,k} + \eta v_k \mathbf{e}_k, \quad (6)$$

where $\mathbf{p}_{R,k}$ is the robot position, $\mathbf{e}_k = (\cos \theta_k, \sin \theta_k)^\top$, and $\eta > 0$ is the center-shift time parameter. With R_k denoting the planar rotation matrix for heading θ_k , define the shape matrix $\Sigma_k = R_k \text{diag}(a_k^2, b_k^2) R_k^\top$. The associated influence region is

$$\mathcal{E}_k = \{\mathbf{p} \in \mathbb{R}^2 \mid (\mathbf{p} - \mathbf{c}_k)^\top \Sigma_k^{-1} (\mathbf{p} - \mathbf{c}_k) \leq 1\}. \quad (7)$$

The sequence $\{\mathcal{E}_k\}_{k=0}^{T-1}$ forms the VRC along the candidate trajectory, retaining the temporal order of the robot's predicted influence regions.

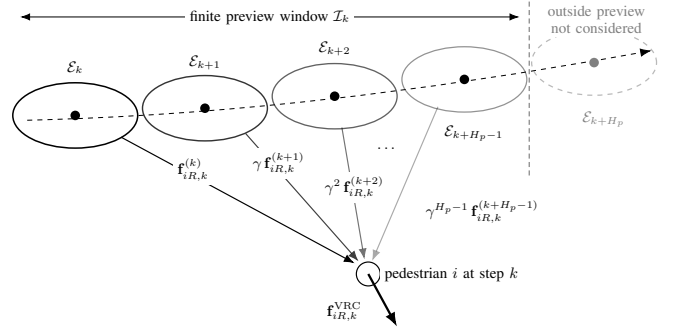


Fig. 2. Finite-preview VRC interaction. The dashed curve is the robot trajectory. Contributions from $\ell \in \mathcal{I}_k$ are evaluated at $\mathbf{p}_{i,k}$, temporally weighted, and normalized to obtain $\mathbf{f}_{iR,k}^{\text{VRC}}$.

B. VRC-Induced Interaction Force

Motivated by social force models [14], [15], the VRC induces a behavioral repulsion rather than a physical contact force. For pedestrian i at predicted position $\mathbf{p}_{i,k}$, let $\mathbf{r}_{i,k}^{(\ell)} = \mathbf{p}_{i,k} - \mathbf{c}_\ell$. The normalized elliptical distance and unit outward normal are

$$\begin{aligned} \rho_{i,k}^{(\ell)} &= \sqrt{(\mathbf{r}_{i,k}^{(\ell)})^\top \Sigma_\ell^{-1} \mathbf{r}_{i,k}^{(\ell)}}, \\ \mathbf{n}_{i,k}^{(\ell)} &= \frac{\Sigma_\ell^{-1} \mathbf{r}_{i,k}^{(\ell)}}{\|\Sigma_\ell^{-1} \mathbf{r}_{i,k}^{(\ell)}\|_2}. \end{aligned} \quad (8)$$

The force follows this ellipse normal, not the radial direction from the ellipse center. Using an exponentially decaying interaction [29], [30], we bound the force magnitude and restrict the interaction to a finite neighborhood:

$$s_{i,k}^{(\ell)} = \begin{cases} \min \left\{ f_{\text{max}}, \alpha \exp \left(\frac{1 - \rho_{i,k}^{(\ell)}}{\beta} \right) \right\}, & \rho_{i,k}^{(\ell)} < \rho_{\text{cut}}, \\ 0, & \text{otherwise.} \end{cases} \quad (9)$$

Here, $\alpha, \beta > 0$ control nominal strength and spatial decay, $\rho_{\text{cut}} > 1$ is the influence cutoff, and $f_{\text{max}} > 0$ bounds the force magnitude. The single-ellipse contribution is $\mathbf{f}_{iR,k}^{(\ell)} = s_{i,k}^{(\ell)} \mathbf{n}_{i,k}^{(\ell)}$, where R denotes interaction with the robot.

To model limited anticipation, consider the preview indices $\mathcal{I}_k = \{k, \dots, \min(k + H_p - 1, T - 1)\}$, with $H_p \ll T$. The aggregate force is the temporally weighted average

$$\mathbf{f}_{iR,k}^{\text{VRC}} = \frac{\sum_{\ell \in \mathcal{I}_k} \gamma^{\ell-k} \mathbf{f}_{iR,k}^{(\ell)}}{\sum_{\ell \in \mathcal{I}_k} \gamma^{\ell-k}}, \quad (10)$$

where $0 < \gamma \leq 1$, with $\gamma < 1$ favoring more imminent robot motion. Normalization prevents the force magnitude from scaling merely with the number of preview ellipses. All contributions are evaluated at the same pedestrian position $\mathbf{p}_{i,k}$ before advancing its state, as shown in Fig. 2.

VI. BELIEF-INFORMED CROWD NAVIGATION WITH MPPI

A. Online Responsiveness Inference

Pedestrians may respond with different strengths to the robot's motion, even under similar interaction conditions [3]. To capture this variability, we introduce a latent responsiveness gain $\kappa_{i,t} \in [0, \kappa_{\max}]$ for each pedestrian i , where $\kappa_{\max} > 1$. The gain scales the nominal VRC interaction force defined in (10):

$$\mathbf{f}_{iR,k|t}^{\text{resp}} = \kappa_{i,t} \mathbf{f}_{iR,k|t}^{\text{VRC}}. \quad (11)$$

Here, t indexes replanning cycles and k prediction steps. Zero, unit, and larger gains represent negligible, nominal, and stronger responses, respectively. The gain remains fixed within each rollout, while its belief is updated between cycles. It is an effective coefficient relative to the VRC model, not an intrinsic pedestrian trait.

Relative to a local constant-velocity reference with zero acceleration, the observed acceleration residual is

$$\mathbf{y}_{i,t} = \frac{\mathbf{v}_{i,t}^{\text{obs}} - \mathbf{v}_{i,t-1}^{\text{obs}}}{\Delta t}. \quad (12)$$

This reference is consistent with the predictor's zero relaxation term at the initial observed state (Section VI-B). To associate this observation with the preceding robot decision, we cache

$$\mathbf{b}_{i,t-1} = \mathbf{f}_{iR,0|t-1}^{\text{VRC}}(\mathbf{p}_{i,t-1}^{\text{obs}}; \mathbf{x}_{t-1}^*, \mathbf{u}_{t-1}^*). \quad (13)$$

The feature uses the final selected plan and the same preview length H_p , temporal discount, and normalization as (10). It is paired with $\mathbf{y}_{i,t}$ after executing the first control input. The preview approximates robot motion anticipated by the pedestrian; its unexecuted tail is not treated as an observed trajectory.

We adopt the observation likelihood

$$p(\mathbf{y}_{i,t} | \kappa_{i,t}) = \mathcal{N}(\mathbf{y}_{i,t}; \kappa_{i,t} \mathbf{b}_{i,t-1}, \Sigma_e), \quad (14)$$

where $\Sigma_e \succ 0$ approximates measurement noise and unmodeled accelerations from intent changes and pedestrian interactions. The shared VRC construction maintains consistent gain scaling between inference and prediction, although systematic model mismatch can still bias the estimate.

Following (1), each belief $q_{i,t}(\kappa_i)$ is updated from its predictive prior $q_{i,t}^-$ using (14). For online computation, we use a Gaussian assumed-density approximation with location $\mu_{i,t}$ and variance parameter $P_{i,t}$, initialized at (μ_0, P_0) for each new pedestrian. The predictive parameters are $\mu_{i,t}^- = \mu_{i,t-1}$ and $P_{i,t}^- = P_{i,t-1} + Q_\kappa$, where $Q_\kappa \geq 0$ allows adaptation to gradual behavioral changes.

Define the scalar information score

$$h_{i,t} = \mathbf{b}_{i,t-1}^\top \Sigma_e^{-1} \mathbf{b}_{i,t-1}. \quad (15)$$

When $h_{i,t} \geq h_{\min}$, the unconstrained linear-Gaussian update is

$$\begin{aligned} P_{i,t} &= [(P_{i,t}^-)^{-1} + h_{i,t}]^{-1}, \\ \tilde{\mu}_{i,t} &= P_{i,t} \left[\frac{\mu_{i,t}^-}{P_{i,t}^-} + \mathbf{b}_{i,t-1}^\top \Sigma_e^{-1} \mathbf{y}_{i,t} \right]. \end{aligned} \quad (16)$$

We constrain the location using $\mu_{i,t} = \Pi_\kappa(\tilde{\mu}_{i,t})$, with $\Pi_\kappa(z) = \min\{\kappa_{\max}, \max\{0, z\}\}$. For $h_{i,t} < h_{\min}$, the measurement update is skipped, giving $(\mu_{i,t}, P_{i,t}) = (\mu_{i,t}^-, P_{i,t}^-)$, to avoid attributing motion changes to responsiveness under weak robot influence.

Location projection and bounded sampling during prediction approximate bounded Bayesian inference rather than an exact truncated-Gaussian update. Informative observations reduce $P_{i,t}$ relative to $P_{i,t}^-$, but $Q_\kappa > 0$ means that uncertainty need not decrease monotonically across cycles.

B. Belief-Conditioned Pedestrian Prediction

Given a candidate robot trajectory and the current responsiveness beliefs, we predict pedestrian motion by combining a local motion reference with the robot-induced response in (11). Since pedestrian goals are unobserved, the current measured velocity $\bar{\mathbf{v}}_{i,t} = \mathbf{v}_{i,t}^{\text{obs}}$ is used as a fixed nominal velocity within each rollout. A relaxation term restores this nominal motion following robot-induced deviations [14].

This approximation avoids explicitly reconstructing pedestrian goals or propagating pedestrian-pedestrian and obstacle interaction forces. These effects are partially reflected in the current measured velocity, but their future evolution is not explicitly modeled. The nominal velocity and initial state are refreshed at every replanning cycle.

The individual responsiveness beliefs are approximated as conditionally independent given the observation history:

$$q_t(\boldsymbol{\kappa}) = \prod_{i=1}^N q_{i,t}(\kappa_i), \quad \boldsymbol{\kappa} = (\kappa_1, \dots, \kappa_N). \quad (17)$$

We draw S joint responsiveness realizations. Each joint sample assigns one responsiveness gain to every pedestrian and keeps these gains fixed over the prediction horizon. With the Gaussian approximation and bounded-gain handling described above, each component is sampled as

$$z_{i,t}^{(s)} \sim \mathcal{N}(\mu_{i,t}, P_{i,t}), \quad \kappa_{i,t}^{(s)} = \Pi_\kappa(z_{i,t}^{(s)}), \quad (18)$$

where $s = 1, \dots, S$. Each sampled gain is held fixed throughout its rollout, representing one consistent response hypothesis over the prediction horizon.

For each realization, the pedestrian state is initialized as $\hat{\mathbf{p}}_{i,0|t}^{(s)} = \mathbf{p}_{i,0}^{\text{obs}}$ and $\hat{\mathbf{v}}_{i,0|t}^{(s)} = \mathbf{v}_{i,0}^{\text{obs}}$. Under the unit-mass convention, the dynamics are discretized using a semi-implicit Euler update:

$$\hat{\mathbf{v}}_{i,k+1|t}^{(s)} = \hat{\mathbf{v}}_{i,k|t}^{(s)} + \Delta t \left[\frac{\bar{\mathbf{v}}_{i,t} - \hat{\mathbf{v}}_{i,k|t}^{(s)}}{\tau_{\text{relax}}} + \kappa_{i,t}^{(s)} \mathbf{f}_{iR,k|t}^{\text{VRC},(s)} \right], \quad (19)$$

$$\hat{\mathbf{p}}_{i,k+1|t}^{(s)} = \hat{\mathbf{p}}_{i,k|t}^{(s)} + \Delta t \hat{\mathbf{v}}_{i,k+1|t}^{(s)}, \quad (20)$$

where $\tau_{\text{relax}} > 0$ is the velocity relaxation time. The VRC force is evaluated at the current sampled pedestrian position using the candidate robot trajectory. Consequently, different robot candidates induce different pedestrian trajectory distributions.

Applying (19) and (20) for $k = 0, \dots, T-1$ produces the sampled crowd trajectories $\hat{\mathbf{X}}_P^{(s)}(\mathbf{x}, \mathbf{u})$. Here, T is the prediction horizon, corresponding to K in the VRC construction. Their empirical distribution is

$$p(\hat{\mathbf{X}}_P | \mathbf{x}, \mathbf{u}, q_t) \approx \frac{1}{S} \sum_{s=1}^S \delta_{\hat{\mathbf{X}}_P^{(s)}(\mathbf{x}, \mathbf{u})}, \quad (21)$$

where $\delta_{\mathbf{x}}$ denotes a point mass at trajectory $\mathbf{x}$.

The same responsiveness samples are reused across all robot candidates and their refinement within a replanning cycle to reduce sampling-induced variability in candidate comparisons. Beliefs remain fixed during these hypothetical rollouts and are updated only from actual observations. The resulting distribution captures uncertainty in the response gain, rather than all sources of pedestrian motion uncertainty.

Fixing all gains to one yields the nominal VRC prediction, while fixing them to zero yields constant-velocity motion under the stated initialization. The belief-conditioned model therefore incorporates both cases within the same prediction formulation.

C. Structured MPPI Candidate Sampling

Conventional MPPI typically samples control perturbations around a single nominal sequence [12]. While efficient for local refinement, such a unimodal sampling distribution may provide insufficient exploration in crowd-navigation problems, where qualitatively different interaction strategies can coexist [31]. When approaching a pedestrian, the robot may pass on either side or adjust its speed to change the timing of the encounter.

To explicitly preserve such multimodality, we construct five structured proposal modes,

$$\mathcal{M} = \{\text{nominal, pass-left, pass-right, yield, pass-first}\}.$$

Each mode is sampled and refined independently, retaining these alternatives until the final selection. This avoids averaging trajectories from distinct modes into undesirable intermediate regions, which can degrade optimization performance [32].

At replanning cycle t , the previously selected control sequence provides a warm start $\mathbf{u}_t^{\text{warm}} = \text{shift}(\mathbf{u}_{t-1}^*)$, where $\text{shift}(\cdot)$ removes the executed first input and repeats the final input to maintain the horizon length. This shifted sequence serves as the common reference for all five modes.

The proposal mean and sampled sequences of mode $m \in \mathcal{M}$ are

$$\begin{aligned} \bar{\mathbf{u}}_t^m &= \mathbf{u}_t^{\text{warm}} + \Delta \mathbf{u}_t^m, \\ \mathbf{u}_t^{m,b} &= \bar{\mathbf{u}}_t^m + \epsilon_t^{m,b}, \end{aligned} \quad (22)$$

where $b = 1, \dots, B$, and $\epsilon_t^{m,b} \sim \mathcal{N}(\mathbf{0}, I_T \otimes \Sigma_u)$. Each mode receives B samples, producing $5B$ robot candidates.

The nominal mode uses $\Delta \mathbf{u}_t^m = \mathbf{0}$. Passing modes apply paired angular-velocity pulses with opposite signs, while yielding and passing-first modes apply negative and positive linear-velocity pulses, respectively. These perturbations explore lateral and timing alternatives while preserving the navigation goal.

Each sampled sequence is propagated from the current observed robot state through (2). Its resulting trajectory defines the VRC and the corresponding belief-conditioned pedestrian predictions used for risk-sensitive evaluation.

D. Risk-Sensitive Trajectory Evaluation and MPPI Update

For compactness, the conditioning time t is omitted below. For responsiveness realization s , define the predicted robot-pedestrian separation

$$d_{i,k}^{(s)} = \|\mathbf{p}_{R,k} - \hat{\mathbf{p}}_{i,k}^{(s)}\|_2. \quad (23)$$

A bounded proximity penalty is

$$\psi(d) = \min\{\exp[-c_d(d - r_{\text{safe}})], 1\}, \quad (24)$$

where $c_d > 0$ controls its spatial decay and $r_{\text{safe}} = r_0$ is the prescribed center-to-center safety distance.

The complete trajectory cost for realization s is

$$\begin{aligned} J^{(s)}(\mathbf{u}) &= \sum_{k=0}^{T-1} w_g \|\mathbf{p}_{R,k+1} - \mathbf{g}\|_2 && \text{(Goal tracking)} \\ &+ \sum_{k=0}^{T-1} w_s \|\mathbf{u}_k - \mathbf{u}_{k-1}\|_2 && \text{(Control smoothness)} \\ &+ \sum_{k=0}^{T-1} w_c (1 + \gamma_c^k) \sum_{i=1}^N \psi(d_{i,k+1}^{(s)}) && \text{(Interaction cost)} \end{aligned} \quad (25)$$

where $\mathbf{g}$ is the robot goal, $w_g, w_s, w_c > 0$ are cost weights, and $0 < \gamma_c \leq 1$. We use $\mathbf{u}_{-1} = \mathbf{0}$. The three terms penalize goal distance along the horizon, control variation, and pedestrian proximity, respectively. Responsiveness uncertainty enters through the sampled pedestrian predictions in the proximity term.

Risk is evaluated across complete trajectory costs. The empirical mean is

$$\hat{\mathbb{E}}[J(\mathbf{u})] = \frac{1}{S} \sum_{s=1}^S J^{(s)}(\mathbf{u}). \quad (26)$$

For confidence level $\zeta \in (0, 1)$, the empirical upper-tail conditional value-at-risk (CVaR) [33] is

$$\widehat{\text{CVaR}}_{\zeta}[J(\mathbf{u})] = \min_{\xi \in \mathbb{R}} \left\{ \xi + \frac{\sum_{s=1}^S [J^{(s)}(\mathbf{u}) - \xi]_+}{(1 - \zeta)S} \right\}, \quad (27)$$

where $[z]_+ = \max\{z, 0\}$. CVaR emphasizes the upper tail of the trajectory-cost distribution induced by the sampled pedestrian responses.

The planning objective combines mean and tail risk:

$$\mathcal{J}(\mathbf{u}) = (1 - \chi) \hat{\mathbb{E}}[J(\mathbf{u})] + \chi \widehat{\text{CVaR}}_{\zeta}[J(\mathbf{u})], \quad \chi \in [0, 1]. \quad (28)$$

The settings $\chi = 0$ and $\chi = 1$ recover mean-only and CVaR-only evaluation, respectively. Intermediate values balance expected performance and adverse outcomes. All temporal and pedestrian costs are accumulated before risk aggregation, so

the tail corresponds to coherent crowd-response realizations over the full horizon.

Within each mode, MPPI combines the sampled control sequences using weights derived from $\mathcal{J}$, producing one refined sequence $\tilde{\mathbf{u}}_t^m$ [34].

Because refinement changes the robot trajectory and its predicted influence on pedestrians, each refined sequence is evaluated again: the robot trajectory is propagated, the VRC and pedestrian predictions are recomputed, and $\mathcal{J}$ is recalculated using the same responsiveness samples as in the initial evaluation. The final sequence is selected by

$$m_t^* = \arg \min_m \mathcal{J}(\tilde{\mathbf{u}}_t^m), \quad \mathbf{u}_t^* = \tilde{\mathbf{u}}_t^{m_t^*}. \quad (29)$$

This completes the receding-horizon loop between interaction observation, belief update, pedestrian prediction, and robot planning. The complete algorithm is summarized in Algorithm 1.

VII. EXPERIMENTS

A. Simulation Setup

We evaluate BIRA-MPPI in SocNavGym [35], which supports closed-loop navigation with reactive pedestrians. The environment is a $12\text{m} \times 12\text{m}$ square centered at the origin, containing $N = 10$ dynamic pedestrians and no static obstacles or physical walls. The robot navigates from $(-4, -4)$ to $(4, 4)$. For each pedestrian policy, we generate 50 seeded Monte Carlo scenarios by randomizing pedestrian initial and goal positions, and reuse identical seeds across all compared methods.

Pedestrian policies: All pedestrians are simulated using ORCA [16], with three robot-response variants: **Probabilistic ORCA:** pedestrians react to the robot probabilistically, representing uncertain and moderately cooperative behavior. **Uncooperative ORCA:** pedestrians exclude the robot from their collision-avoidance response and therefore do not actively yield to it. **Cooperative ORCA:** pedestrians consistently account for the robot and exhibit stronger collision-avoidance responses.

Compared methods: We compare three variants sharing the same structured-MPPI planner, robot dynamics and observations, differing only in pedestrian prediction and uncertainty treatment: **Constant velocity (CV)-MPPI:** non-interactive constant-velocity prediction. **VRC-MPPI:** candidate-conditioned prediction with fixed responsiveness $\kappa_i = 1$. **BIRA-MPPI:** online responsiveness inference with the proposed mean-CVaR objective.

Implementation: All methods replan every $\Delta t = 0.1\text{s}$ over a horizon of $H = 40$ steps (4s). At each planning step, 900 robot rollouts are distributed equally across five independently refined trajectory modes. The MPPI temperature is $\lambda = 0.1$, with control-noise standard deviations $\sigma_v = 0.1\text{m/s}$ and $\sigma_\omega = 0.35\text{rad/s}$. For uncertainty-aware evaluation, BIRA-MPPI draws $S = 16$ joint responsiveness samples per replanning cycle and reuses them across all robot candidates and their refined trajectories. The Gaussian parameters are initialized as $(\mu_0, P_0) = (1, 0.6^2)$. Responsiveness samples are clipped to $[0, \kappa_{\max}]$, with $\kappa_{\max} = 2$,

Algorithm 1 BIRA-MPPI

Input: robot state $\mathbf{x}_{R,t}$, pedestrian observations $\mathbf{X}_{P,t}^{\text{obs}}$, goal $\mathbf{g}$
Given: mode set $\mathcal{M}$, B samples per mode, S belief samples, noise covariance Σ_u , temperature λ
Memory: previous belief q_{t-1} , selected plan $\mathbf{u}_{t-1}^*$, and interaction history

- 1: **Step 1: Update responsiveness belief**
- 2: Update q_t using observed motion and cached VRC interaction features
- 3: Draw shared bounded gain samples $\mathcal{K}_t = \{\kappa_t^{(s)}\}_{s=1}^S$ using (18)
- 4: **Step 2: Generate structured candidates**
- 5: Construct the five mode means $\{\bar{\mathbf{u}}_t^m\}_{m \in \mathcal{M}}$ from the time-shifted plan $\mathbf{u}_{t-1}^*$
- 6: Sample B admissible control sequences $\{\mathbf{u}_t^{m,b}\}_{b=1}^B$ around each $\bar{\mathbf{u}}_t^m$ using Σ_u
- 7: **Step 3: Evaluate and refine each mode**
- 8: **for** $m \in \mathcal{M}$ **do**
- 9: **for** $b = 1, \dots, B$ **do**
- 10: Roll out $\mathbf{u}_t^{m,b}$ and construct its VRC
- 11: Predict pedestrian trajectories conditioned on this candidate and each gain sample in $\mathcal{K}_t$
- 12: Compute sampled costs $\{J_t^{(s)}(\mathbf{u}_t^{m,b})\}_{s=1}^S$ and aggregate $\mathcal{J}(\mathbf{u}_t^{m,b}; q_t)$ using (28)
- 13: **end for**
- 14: Obtain $\tilde{\mathbf{u}}_t^m$ through an MPPI update within mode m , using these costs and λ
- 15: **end for**
- 16: **Step 4: Re-evaluate and select**
- 17: Recompute robot and VRC-conditioned pedestrian predictions for each $\tilde{\mathbf{u}}_t^m$ using the same $\mathcal{K}_t$, and re-evaluate $\mathcal{J}(\tilde{\mathbf{u}}_t^m; q_t)$
- 18: $m_t^* \leftarrow \arg \min_{m \in \mathcal{M}} \mathcal{J}(\tilde{\mathbf{u}}_t^m; q_t)$
- 19: $\mathbf{u}_t^* \leftarrow \tilde{\mathbf{u}}_t^{m_t^*}$
- 20: Retain $\mathbf{u}_t^*$ and its VRC preview features for the next planning and belief updates
- 21: **return** the first control input $\mathbf{u}_{0|t}^*$

using (18). All trajectory rollouts and cost evaluations are parallelized in PyTorch on an NVIDIA GeForce RTX 5060 Laptop GPU.

Performance metrics: **Success rate:** percentage of episodes reaching the goal successfully. **Collision rate:** percentage of episodes containing a collision. **Navigation time:** mean simulated time to reach the goal over successful episodes. **Path length:** mean cumulative Euclidean distance traveled by the robot over successful episodes. **Computational time:** mean planning time per decision.

B. Results and Discussion

Fig. 3 illustrates a closed-loop BIRA-MPPI episode. As nearby pedestrians move across the robot's route, the selected plan and its VRC are updated to accommodate the changing interaction geometry. The robot maneuvers through the cen-

TABLE I
SIMULATION RESULTS

Pedestrian Behavior	Predictor	Success Rate $\uparrow$	Collision Rate $\downarrow$	Navigation Time $\downarrow$	Path Length $\downarrow$	Computational Time $\downarrow$
Uncooperative	CV-MPPI	74	20	23.13	14.97	9.17
	VRC-MPPI	50	36	23.09	13.71	31.56
	BIRA-MPPI	82	14	23.32	15.14	57.98
Probabilistic	CV-MPPI	74	12	22.77	14.59	9.13
	VRC-MPPI	68	16	25.06	13.65	31.60
	BIRA-MPPI	88	4	22.70	15.11	57.95
Cooperative	CV-MPPI	84	6	23.91	14.92	9.13
	VRC-MPPI	72	12	22.84	13.46	31.61
	BIRA-MPPI	86	4	23.95	15.08	58.06

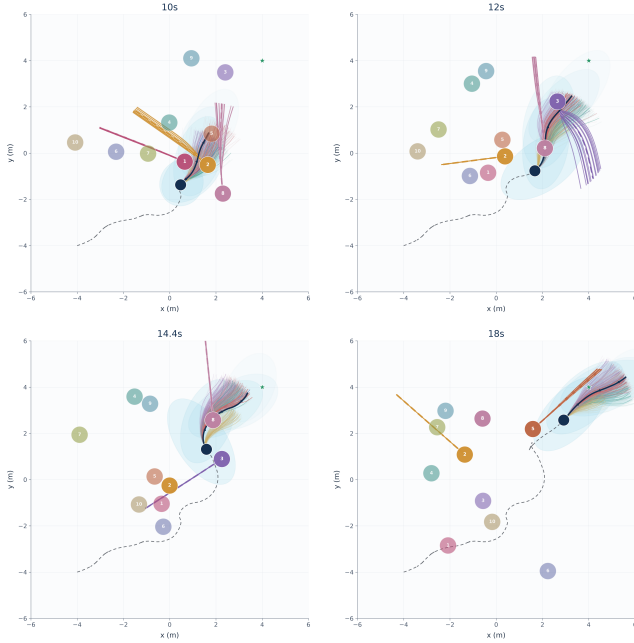


Fig. 3. Four chronological snapshots of BIRA-MPPI navigating among ten pedestrians. The dark-blue disk and green star denote the robot and its goal; numbered disks identify pedestrians. The dashed gray line shows the executed robot path, thin lines originating from the robot show candidate trajectories, and the thick dark-blue line indicates the selected plan. Translucent cyan ellipses visualize the VRC, while lines matching pedestrian colors show sampled pedestrian predictions.

tral crowd and subsequently progresses toward the goal. The snapshots illustrate how distributional pedestrian prediction is integrated with repeated trajectory replanning.

Table I shows that BIRA-MPPI achieves the highest observed success rate and the lowest collision rate across all three pedestrian behaviors. The results highlight three main findings.

Takeaway 1: The largest gains occur under unreliable cooperation. BIRA-MPPI improves most over CV-MPPI in the probabilistic setting, increasing success by 14 percentage points while reducing collisions from 12% to 4%. Its smaller advantage in the cooperative setting suggests that accounting for uncertain responses is particularly use-

ful when pedestrian avoidance cannot be reliably assumed. In contrast, fixed-response VRC-MPPI underperforms CV-MPPI across all settings, indicating that explicitly modeling interactions does not by itself ensure better navigation. A plausible explanation is a mismatch between assumed and actual pedestrian responses. These comparisons support the combined belief-aware and risk-sensitive design, but do not isolate the contribution of either component.

Takeaway 2: Higher reliability entails a modest traversal-efficiency trade-off. BIRA-MPPI produces slightly longer paths than CV-MPPI while maintaining comparable navigation times. The shorter paths of VRC-MPPI, however, accompany substantially poorer navigation reliability and should not be interpreted as superior overall performance. Importantly, path length and navigation time are averaged over successful episodes, so the compared planners are evaluated on different episode subsets. These metrics therefore describe successful traversal efficiency rather than an unconditional performance advantage.

Takeaway 3: BIRA-MPPI increases computation while retaining real-time feasibility. The additional computational cost arises from belief-aware distributional prediction and risk-sensitive trajectory evaluation. Nevertheless, the computation remains below the 100 ms simulation step, allowing BIRA-MPPI to operate within the 10 Hz replanning budget. Thus, the improved navigation reliability is achieved at increased, but practically acceptable, computational cost.

VIII. CONCLUSION

We presented BIRA-MPPI, which integrates VRC-based interaction modeling, online variational Bayesian responsiveness inference, and risk-sensitive MPPI. Pedestrian-specific beliefs generate trajectory distributions conditioned on each robot candidate, supporting mean-CVaR evaluation and independent motion-mode refinement. Across three ORCA response regimes in SocNavGym, BIRA-MPPI achieves higher success and lower collision rates than constant-velocity and fixed-responsiveness baselines, with a mean planning time of approximately 58 ms. These results support the value of adapting pedestrian response estimates while retaining their uncertainty during planning.

Limitations and future work. The modeled uncertainty is restricted to scalar responsiveness gains; unmodeled intent changes and pedestrian–pedestrian interactions may bias inference and prediction. Evaluation is limited to obstacle-free, ORCA-based simulations. Future work will investigate richer interaction models, component-wise ablations, and real-robot validation in more complex environments.

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